

Quotient Laws of ExponentsQuestion Bank 3.02 **ANSWER KEY**

Show your work. Circle your final answer.

For questions 1 through 3, show the quotient of powers law, then evaluate.

1. $\frac{10^5}{10^3}$

$$10^{5-3} = 10^2 = 100$$

2. $\frac{8^4}{8^1}$

$$8^{4-1} = 8^3 = 512$$

3. $\frac{(-6)^6}{(-6)^5}$

$$(-6)^{6-5} = (-6)^1 = -6$$

One negative factor left.

4. When dividing two powers that have the same nonzero base, keep the base and _____ the exponents.

subtract

A cloze blank: the word is the answer.

For questions 5 and 6, apply the power of a quotient law. Do not simplify the fraction.

5. $\left(\frac{-5}{10}\right)^7$

$$\frac{(-5)^7}{10^7}$$

Equivalent to $(-0.5)^7 = -0.0078125$, but the rewrite is what is asked.

6. $\left(\frac{3}{-4}\right)^5$

$$\frac{3^5}{(-4)^5}$$

Keep the parentheses on the negative base.

The zero exponent.

7. Fill in the missing value in the pattern: $4^4 = 256$, $4^3 = 64$, $4^2 = 16$, $4^1 = 4$,
 $4^0 =$ _____

1

Each value is the previous one divided by 4; $4 \div 4 = 1$.

8. Evaluate 9^0 and explain in one sentence why your answer is not 0.

1 — the exponent counts factors of 9; with none multiplied in, what is left is 1 (or: $9^3/9^3 = 1$ and $9^3/9^3 = 9^0$).

For questions 9 through 11, rewrite the expression using a law of exponents, name the law, then evaluate.

9. $\frac{7.6^5}{7.6^3}$

$7.6^2 = 57.76$ — quotient of powers

10. $\left(\frac{10}{11}\right)^0$

1 — zero exponent

11. $\left(\frac{0.8}{1.6}\right)^4$

$\frac{0.8^4}{1.6^4} = \frac{0.4096}{6.5536} = 0.0625$ — power of a quotient (or $0.5^4 = 0.0625$)

Working backwards.

12. Find the value of n that makes each statement true.

a. $\frac{3^7}{3^n} = 3^2$

$n = 5$

$7 - n = 2.$

b. $\left(\frac{5}{2}\right)^n = \frac{125}{8}$

$n = 3$

$5^3 = 125$ and $2^3 = 8.$

c. $\frac{6^n}{6^4} = 1$

$n = 4$

The quotient is 6^0 only when the exponents match.

13. Which expression is equivalent to $\frac{12^9}{12^3}$?

- ☐ A. 1^6
- ☒ B. 12^6
- ☐ C. 12^3
- ☐ D. 12^{12}

B

Same base, subtract.

14. Select ALL of the expressions whose value is 1.

- ☐ A. $3^0 \cdot 3$
- ☒ B. 5^0
- ☒ C. 1^5
- ☒ D. $\frac{3^4}{3^4}$
- ☒ E. $(-4)^0$
- ☐ F. $\frac{3^4}{3^3}$

B, C, D and E